



INTERN REPORT PHISHING EMAIL DETECTION & AWARENESS REPORT

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Confidentiality Statement

This report contains information related to a phishing email analysis performed during an internship project. The contents of this report are intended for authorized review and should not be distributed or disclosed without appropriate permission.

Disclaimer

This phishing email analysis was conducted as part of a cybersecurity internship project. The findings and observations presented in this report are intended solely for educational and learning purposes.

Assessment Overview

The purpose of this assessment was to analyze a phishing email sample and identify indicators that may be used to deceive recipients. The review focused on examining the sender information, email content, embedded links, attachments, and social engineering techniques used within the message.

The assessment aimed to determine whether the email exhibited characteristics commonly associated with phishing attacks and to provide recommendations for identifying and mitigating similar threats in the future.

Scope

Assessment	Details
Phishing Email Analysis Provided phishing email sample	Scope Sender details, email content, links, attachments, and phishing indicators

Scope Exclusions

Inscope

- ☐ Analysis of one phishing email sample
- ☐ Identification of common phishing indicators
- ☐ Focus on sender details, email content, and links
- ☐ Report prepared only for educational and awareness purposes.

Executive Summary

Phishing is a common cyberattack where attackers send fake emails to trick users into sharing sensitive information such as passwords, OTPs, or personal details. These emails are designed to look real and usually create fear or urgency so that users act without thinking.

In this report, a suspicious email sample has been analyzed to understand how phishing attacks work. The analysis includes checking the email header, sender details, message content, and the link provided in the email using email analysis tools. The email failed important security checks like SPF, DKIM, and DMARC, which indicates that the sender was not genuine.

Several phishing indicators were identified, including a fake sender domain, urgent warning messages, a suspicious verification link, and a generic greeting. Based on these findings, the email was classified as High-Risk Phishing. This report also explains the phishing attack in simple terms and provides basic safety tips to help users identify and avoid phishing emails in the future.

FINDINGS SUMMARY

:

Step	Action	Recommendation
1	Sender Address Spoofing / Email Impersonation	Verify sender domains and implement SPF, DKIM, and DMARC controls.
2	Social Engineering and Urgency Tactics	Avoid acting on urgent requests without independent verification. data minimization principles.
3	Malicious PDF Attachment with Embedded Link	Do not open unsolicited attachments and scan files before use.

INTRODUCTION

Phishing is a type of cyber attack where attackers send fake emails that look like they are from trusted sources. These emails are used to trick users into clicking harmful links or sharing sensitive information such as passwords or OTPs.

Phishing attacks are a serious problem because they mainly target users who are not aware of how to identify fake emails. Messages that create urgency, such as account warnings or security alerts, often confuse users and lead to mistakes.

This report focuses on understanding phishing emails by analyzing a sample email and identifying common phishing signs. The goal of this report is to improve user awareness and help users stay safe from phishing attacks.

Analysis of Sample Emails

Spoofed Email Address

Description:	The email attempted to impersonate PayPal by displaying a trusted sender name. However, analysis revealed that the actual sender domain did not match the displayed identity, indicating a spoofing attempt designed to deceive recipients.
Impact:	High
Recommendation:	Verify sender domains before interacting with emails and implement SPF, DKIM, and DMARC controls to reduce email spoofing.

Proof of Concept

The phishing email sample was manually reviewed to identify indicators of malicious intent.

Evidence observed during the analysis included:

- Mismatch between the displayed sender identity and the actual sender email address.
- Use of an attention-grabbing subject line designed to create urgency.

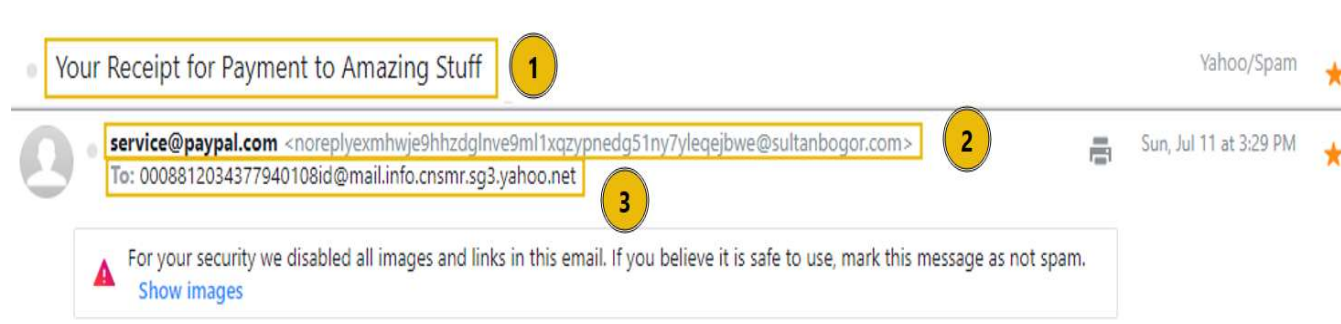


Figure 1: Email Header Analysis

- Presence of a shortened URL embedded within the call-to-action button. (Taking a look at the email body below, we can see that this is a receipt for a purchase of gift cards. The email is designed to appear as a legitimate email from PayPal. There aren't any attachments associated with this email, and the only interactive element in this email is the Cancel the order button)
- By inspecting the raw source of the email, we can further investigate the hyperlinks and underlying HTML that form the message. The Cancel the order button leads to the shortened URL. Because the attacker is using a URL shortening service, the final destination is obfuscated, making it impossible to verify the landing page at a glance. As a rule of thumb, you should never interact with buttons or links without first confirming exactly where they lead.

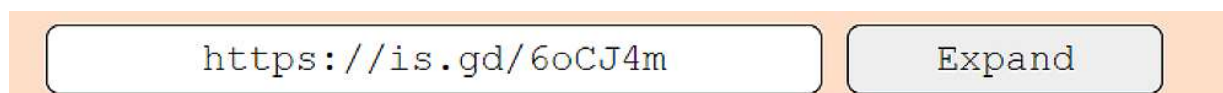

```

<!-- CTA text 17/21 -->
<td bgcolor="#0070BA" align="center" valign="middle" style="padding:13px
90px 13px 90px; display: block; color:#ffffff !important; text-decoration:none; white-space: nowrap;
-webkit-print-color-adjust: exact; display: block; font-family:Calibri, Trebuchet, Arial, sans serif;
font-size:17px; line-height:21px; color:#ffffff !important;border-radius:25px;" class="ppsans
mobilePadding9"><a type="Link" isLinkToBeTracked="true" target="_blank"
linkId="e854f9fff3d47fb841257f9bebb4d1b6" style="text-decoration:none;color:#ffffff !important;
white-space: nowrap;" href="https://is.gd/6oCJ4m?#-u7g?ah7Je1";Cancel the order</a></td>
<!-- end CTA text --></tr>

```

Figure 2: Short link behind button

- [WhereGoes](#) that can be used to investigate shortened URLs without having to actually visit the destination. It confirmed this mail is spoofed.



Screenshot of the distant page behind your short URL



Figure 3: Spoofed address found

Sender Address Spoofing

Description:	The email impersonated a trusted distribution center by using a misleading display name. Analysis revealed that the displayed sender identity did not match the actual sender email address, indicating an attempt to deceive the recipient and gain trust.
Impact:	High
Recommendation:	Verify sender email domains before interacting with unsolicited emails and implement SPF, DKIM, and DMARC controls.

Proof of Concept

The email displayed the sender name "Distribution Center" while the actual sender address was identified as "contact@beginpro.club".

Let's take a look at the email below and try to highlight our initial observations.

1. **Subject line:** The subject line uses a fake tracking number to create a sense of urgency, prompting the recipient to click to see their package status
2. **From address:** This is an immediate red flag, as the display name **Distribution Center** does not match the actual sender address contact@beginpro.club
3. **Hyperlink:** The link in the email body matches the subject line, although we do not know where it directs to just yet

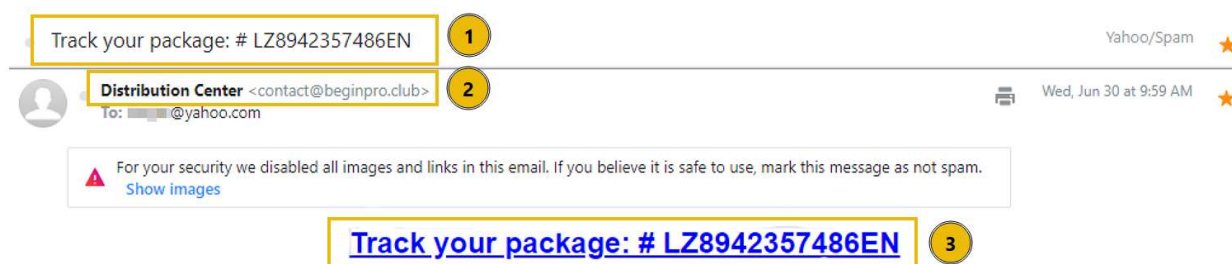


Figure 4: Initial observations

Let's check out the source of the email message to get a closer look at the tracking number hyperlink from above. Here is an image file named Tracking.png. These trackers send information back to the spammer's server. There are several reasons spammers embed [tracking pixels \(opens in new tab\)](#) (very small images) into their emails, and now we can understand why Yahoo automatically blocked the images in this email. Many email providers do the same.

```

<html>
<body>
  <center>
    <a style="color:#000000" href="http://devret.xyz/4833mt11254939vf6888zq22032si1269du1508rr" >
      <font color=indianred size=5 face=Rockwell> <FONT color="#0000FF" face="arial" size=5 ><strong>
    </FONT><br/><br/>
    </a>
    <br />
    
    <map name="grmk"><br />
      <area shape="rect" coords="0, 0, 1500, 1400" href="http://devret.xyz/4833fx11254939ea6888wk22032mk1269ep1508rr" >

    </map>
    <br />
    
    <map name="unsub">
      <area shape="rect" coords="0, 0, 1500, 1400" href="http://devret.xyz/4833jo11254939iz6888xo22032gu1269jm1508uu" >
    </map>
    <br />
  </center>
</body>
</html>



```

Hyperlink Destination

Display Text

Tracking Pixel

Track your package: # LZ8942357486EN

Figure 5: Spoofed destination

Malicious Attachment Containing Embedded Link

Description:	The API does not appear to restrict the number of requests that can be made within a specific time frame. Without rate limiting, APIs can be abused through automated requests, leading to service abuse or performance issues.
Impact:	High
Recommendation:	The email used a PDF attachment containing an embedded hyperlink rather than placing the link directly in the email body. This technique can conceal malicious destinations and bypass simple email filtering mechanisms.

Proof of Concept

The attached PDF contained an "Update Payment Account" link directing users to a non-Netflix domain.

Taking a look at the email body, we can see that the user is informed that there is an issue with their billing information. In order to update their account, they must open the attached PDF file. The attachment contains an embedded link titled Update Payment Account which directs to a URL not associated with a legitimate Netflix domain. We will investigate this email in detail in the next room of this module, [Analysis Tools](#). There are a couple other of points of interest worth noting.

1. The use of an atypical phone number format is a red flag
2. Using a legitimate Netflix help center domain to build a false sense of trust

Please update your payment details

Hello Customer,

We're having some trouble with your current billing information. Please open file (PDF) to update your account.

Need Help? We're here if you need it. Visit the [Help Center](#) or contact us now.

-Your friends at Netflix

Quickstart Call: 801-603-3713 1

Please do not reply to this email, as we are unable to respond from this email address. If you need help or would like to contact us, please visit our [Help Center](#) at [help.netflix.com](#).

2

Email Body

Update Payment Account

If you did not ask to reset your password you may want to review your [recent account access](#) for any unusual activity.

We're here to help if you need it. Visit the [Help Center](#) for more info or [contact us](#).

- Your friends at Netflix

PDF Attachment



Figure 6: Red flags

